PORTFOLIO PROJECT REPORT • GEOSPATIAL ANALYTICS

GeoPulse

Store Catchment & Mobility Analytics for Lagos

A location-intelligence case study exploring where simulated mobility activity concentrates, how store catchments overlap, and what those patterns may mean for retail site planning.

10M	213,256	8
Synthetic GPS pings	Matched ping records	Dashboard pages

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GitHub portfolio project • Reported findings are based on synthetic data and project outputs supplied for this report.

Important: this is a portfolio simulation, not a measurement of real customer visits or actual store performance.

1. Executive Summary

GeoPulse is a retail location-intelligence portfolio project designed to help a coffee-chain real-estate team examine simulated mobility patterns around stores in Lagos. It combines synthetic GPS pings, store catchment matching, spatial summaries, and a multi-page Streamlit dashboard to turn location records into a more decision-oriented view of activity concentration and overlap.

The reported analysis covers 10,000,000 synthetic GPS pings. A derived **store_catchments** table contains 213,256 matched ping records within modelled 500-metre catchments. These records represent ping-to-store matches—not unique visits, people, transactions, or verified footfall.

The project is intended to demonstrate a reproducible analytical workflow and communicate how geospatial evidence could support early-stage site screening. Any real investment decision would require validation with observed footfall, sales, access, rents, local demographics, and competitor data.

Project objectives

- Summarise where simulated mobility activity is concentrated across the modelled store network.
- Explore the spatial overlap between store catchments and the possible implications for site selection.
- Identify hourly activity patterns and spatial density concentrations.
- Present the findings through a dashboard designed for executive review and exploration.

2. Business Problem

Retail expansion teams can be misled by static demographic snapshots when they need to understand how movement patterns vary by place and time. GeoPulse frames a practical question: **where should a retailer investigate next, and what potential overlap should be checked before committing to a new location?**

The case study imagines a coffee-chain real-estate director evaluating potential locations. The dashboard offers descriptive evidence about simulated activity and catchment geometry. It does not establish causal customer switching or prove that one store is cannibalising another.

Questions explored

- Which areas and stores account for the largest share of matched pings?
- When do matched pings peak during the day?
- Where do modelled store catchments overlap?
- How concentrated is activity across the H3 hexagonal grid?

3. Data and Analytical Approach

Dataset / output	Reported size	Interpretation
Synthetic GPS pings	10,000,000 rows	Simulated location observations; not actual customer visits.
Spatially enriched GPS pings	10,000,000 rows	Ping records prepared for geospatial analysis.
store_catchments	213,256 records	Ping-to-store matches within modelled 500 m catchments. A ping may match more than one store.

Workflow overview

1. **Generate / source data** — create a large synthetic GPS-ping dataset for repeatable analysis.
2. **Load and query** — use BigQuery to store and query the dataset and derived store-catchment records.
3. **Enrich spatially** — associate pings with modelled store catchments and calculate summary metrics.
4. **Summarise and visualise** — prepare catchment, time-of-day, store and H3 density views.
5. **Communicate findings** — use a Streamlit dashboard with maps and charts to support exploration.

Technology stack

- **Python and Streamlit** for dashboard application and analytical workflow.
- **Google BigQuery** for cloud data storage and SQL analysis.
- **Folium** for interactive map presentation.
- **Plotly** for interactive charts and hourly patterns.
- **H3 / hexagonal aggregation** for spatial density exploration.

The architecture above reflects the project design and dashboard context available for this report. It should not be read as confirmation that every planned component is production-deployed or fully automated.

4. Dashboard Structure

Dashboard page	Purpose
Executive Dashboard	High-level summary of the main reported metrics and patterns.
Spatial Mobility Map Analysis	Geographic view of simulated mobility and store relationships.
H3 Hexagon Density	Aggregated spatial concentration using hexagonal cells.
3D Hexagon View	A 3D-style spatial density exploration.
Hourly Activity Patterns	Time-of-day distribution of matched activity.
Store and Catchment Analysis	Comparison of store-level counts and modelled catchments.
Final Candidate Recommendation	Illustrative candidate scoring to demonstrate a decision-support workflow.
Contact the Author	Project and author contact information.

5. Reported Findings

The values below are reported project results. They have been carried into this report as supplied and have not been independently recalculated from the underlying dataset.

Activity concentration by area

Victoria Island accounts for 70,036 matched pings (reported as 32.8%). The top three stores account for 88.3% of the 213,256 matched ping records. Ikeja is reported at 1.1%.

Catchment overlap

Lekki catchments are reported to overlap by 81.6%. A total of 44,564 devices appear in both Lekki catchments, representing 84% of devices associated with Lekki A in the reported summary.

Hourly pattern

Activity is reported to peak around 08:00 and 18:00 West Africa Time (WAT); together, those two hours account for approximately 17% of activity.

Ping-to-store distance

The reported average ping-to-store distance is approximately 331–341 metres.

H3 density concentration

At H3 resolution 9, 11 of 69 hexagons are reported to contain half of the matched pings.

How to interpret these results

- A high matched-ping share indicates concentration in the simulated dataset; it does not automatically mean higher sales or more customers.
- Catchment overlap is a geometric / modelled relationship. Device overlap does not by itself prove customer switching or revenue cannibalisation.
- The hourly profile can help form hypotheses about commuter-related demand, but it needs validation against real visits and transactions.
- H3 concentration highlights where the simulated records cluster; it is not a direct measure of population or store performance.

6. Business Implications and Recommended Actions

Use GeoPulse as a screening tool

Use the dashboard to shortlist areas for deeper investigation, not as a standalone location approval mechanism.

Validate the busiest areas

Compare the simulated concentration pattern with measured pedestrian counts, store transactions, and time-stamped sales.

Investigate Lekki overlap

Review the overlap result as a prompt for fieldwork: assess actual travel routes, barriers, access points, and whether the stores serve the same customer missions.

Test candidate sites consistently

Apply the same time windows, catchment definitions, and scoring rules to each candidate. Document the assumptions and sensitivity to alternative radii.

Add commercial context

Before a real decision, incorporate rent, operating costs, local demographics, competitor locations, road access, parking, transit, safety, and sales potential.

7. Limitations and Responsible Use

GeoPulse is currently a synthetic-data portfolio simulation. Its outputs demonstrate analytical methods and dashboard communication, not verified facts about the Lagos retail market.

- **Synthetic observations:** the pings are simulated and should not be presented as actual customer or pedestrian behaviour.
- **Ping versus visit:** repeated pings from a device do not equal distinct visits, customers, or transactions.
- **Fixed catchment radius:** 500 m circles are a simplifying assumption and may not reflect real walking routes, road barriers, or travel time.
- **Overlap is not causation:** overlapping catchments or shared device IDs do not prove cannibalisation.
- **Illustrative candidate score:** the recommendation page uses a modelled score and should not be represented as a validated investment recommendation.
- **Data and pipeline status:** this report describes the reported project workflow. It does not certify that scheduled pipelines, tests, or deployment are operational.

8. Suggested Next Steps

For the next portfolio iteration, prioritise work that improves reproducibility and makes the evidence easier to audit:

- Publish this report and link it from the repository README once reviewed.
- Add a short dashboard preview GIF or walkthrough video so visitors can understand the dashboard before opening it.
- Document dataset generation, schema, catchment logic, and metric definitions.
- Add validation checks for row counts, nulls, timestamp ranges, distance calculations, and duplicate handling.
- Add unit or integration tests for the most important transformations and summary metrics.
- Clearly separate simulated data, observed data, model assumptions, and recommendations in the dashboard.
- Only describe the dashboard as live after deployment is completed and the URL is verified.

9. Conclusion

GeoPulse demonstrates how data analytics and geospatial visualisation can be combined to explore retail mobility questions. Its core value as a portfolio project is the end-to-end story: a large synthetic dataset, spatial matching, summary metrics, an interactive dashboard, and business-facing interpretation with explicit caveats.

The most important next step toward real-world use is not simply adding more visuals; it is validating the measures against observed movement and commercial outcomes, and documenting the assumptions so that decision-makers can judge the evidence appropriately.

Project resources

GitHub repository: github.com/lawrenceruth/GeoPulse

Project design plan: docs/project_blueprint.md

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Report note: Metrics and findings are reproduced from the project context supplied for this report. Validate them against the source data before using them in a public case study or real business decision.